

Circularly Polarized Patch Antennas

6.2300 Class Project: Detailed Plan

Matthew Taylor
EECS
tmatthew@mit.edu

Fiona Daly
Physics
fionad@mit.edu

Gonzalo Esparza
EECS
gje@mit.edu

Abstract—In this paper, two circularly polarized patch antenna designs are compared to a linearly polarized antenna as well as each other, all targeting operation within the 2-3 GHz range. The designs are simulated using both the finite difference time domain method and the finite element method. We fabricate these designs on FR-4 board and evaluate the input return loss and axial ratio of circular polarization for each of these devices.

I. INTRODUCTION & MOTIVATIONS

Modern wireless communication systems rely on antennas to transmit and receive electromagnetic (EM) signals, with WiFi and Bluetooth operating in the 2.45 GHz band. A microstrip patch antenna (MPA) consists of a metallic patch printed onto a grounded dielectric substrate. Its structure supports resonant EM nodes and radiates through the fringing fields at its edges [1]. These antennas can be approximated as half-wavelength resonators with radiating slots, allowing simple models to accurately predict resonances and impedance characteristics [2]. The simple structure of the MPA for both modeling and manufacturing has led to its mainstream use in communication systems. Its use in most major cellphones can be seen in Figure 1.

MPAs have several key limitations, including narrow bandwidths and impedance matching issues [1]. Most MPAs produce linearly polarized EM waves, which can lead to a significant degradation in performance when transmitting and receiving antennas are misaligned. Several approaches have been created to address the issues of polarization on MPAs through alternative designs. A common approach is single-feed geometric perturbation, which uses slotting or corner truncation to generate circular polarization [3].

The use of a finite difference time domain (FDTD) solver for modeling microstrip circuits (MC) was previously demonstrated in [4]. However, the authors model absorbing boundaries using the one-dimensional wave equation, which is less performant at absorption and less generalizable than the perfectly matched layer (PML) boundaries [5] imposed in this work. A similar approach of modeling a MPA with an FDTD solver and PML boundaries was taken in [7], however for the application of a fractal antenna. The truncated corner approach to constructing a circularly polarized antenna was also taken in [8], however for the application of satellite communication with maximal gain.

This work applies these principles to three MPA designs targeted to operate at 2.45 GHz. Sections II-A and II-B intro-



Fig. 1. MPAs inside various Cellular Phones [6]

duce the MPA design process. Section II-C describes the use of numerical models to verify and optimize the designs. Section II-D describes the fabrication and measurement processes. Sections IV and V provide the final results, including the measurement of S_{11} and the polarization characteristics.

II. APPROACH

A. Substrate and Microstrip Launch Characterization

Before the circuit design process, we construct a simple microstrip TEM line (width $W = 2.92$ mm and length $l = 9.6$ cm) on the same FR-4 substrate (height $h = 1.6$ mm) which is used for the MPA fabrication process. We measure S_{11} of the microstrip line terminated with a 50Ω load and use the measurement to approximate the FR-4 relative permittivity $\epsilon_{r,\text{sub}}$, FR-4 bulk conductivity σ_{sub} , and parasitic series inductance L_{series} of the SMA connectors placed at both ends of the microstrip line. With more time, the substrate and launch characterization process would benefit greatly from the use of multiple de-embedding structures to model the connector parasitics more accurately.

A sub-optimal, first-pass estimation of the substrate and launch characteristics resulted in the MPA designs discussed below, which have a resonant frequency about 10% below the target of 2.45 GHz. In the simulation results that follow, we use a refined parameter estimation that provides a significantly better match to the calibration microstrip TEM line measurement and MPA measurements. The resulting calibration fit is shown in Fig. 2, where $\epsilon_{r,\text{sub}} = 4.4$, $\sigma_{\text{sub}} = 0.01$ S/m, and $L_{\text{series}} = 1$ nH.

B. Initial Design

We consider three MC designs, which are depicted in Fig. 3. Circuit 1 provides linear polarization aligned with the white

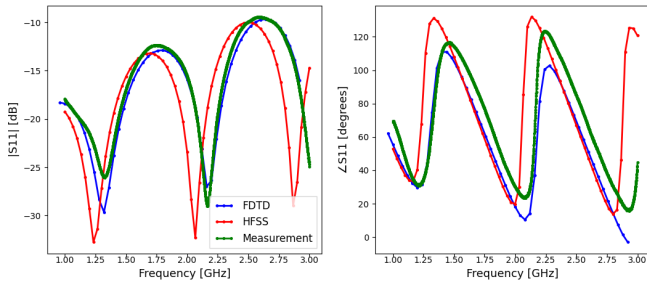


Fig. 2. Measurement and simulation of the S_{11} magnitude and phase for the calibration microstrip TEM line.

arrows in the figure. Circuit 2 provides circular polarization via truncation of two opposite patch corners. The corner truncation results in two resonant modes with orthogonal polarization, and circular polarization ideally results at the midpoint of the two mode frequencies [9]. Circuit 3 provides circular polarization by feeding the patch with a quadrature hybrid coupler, allowing orthogonal modes to be excited with a 90° phase [2].

For all three MC designs, the patch length is initially set to half the effective wavelength in the substrate, $\lambda_{\text{eff}}/2$. The patch length is optimized in simulation to achieve resonance (and minimum axial ratio for Circuit 2 and 3) at the target operating frequency, and the impedance of the patch is simulated to determine the required single-stub matching network. Circuit 3 is not simulated (with exception of the quadrature hybrid coupler in isolation) and is designed such that $L_3 = L_1$, $L_{s3} = L_{s1}$, and $d_{s3} = d_{s1}$. The tuning stubs in Circuit 3 must be placed in close proximity due to limited board area.

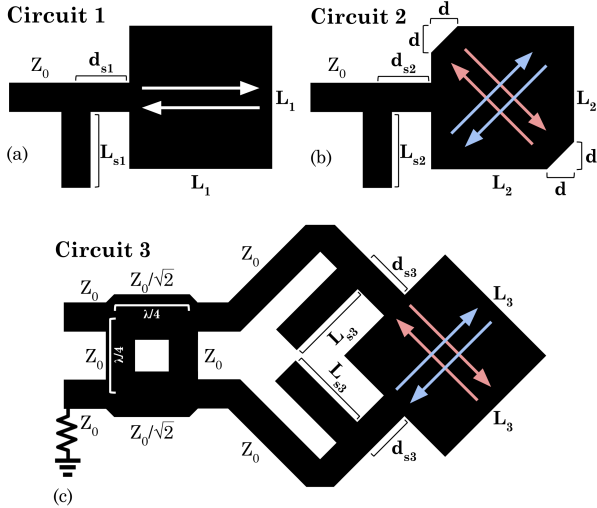


Fig. 3. The three MC designs. (a): Linearly polarized. (b): Circularly polarized with single feed. (c): Circularly polarized with double feed.

C. Simulation and Optimization

To optimize the circuit designs discussed above, we simulate the designs using a custom FDTD simulator, which is verified

against the Ansys HFSS finite-element method (FEM) simulator. The FDTD simulator implements convolutional CFS-PML absorbing boundaries [10] and provides conformal treatment of geometries that do not perfectly sit on the grid [11]. S_{11} is extracted as follows. First, the frequency-dependent impedance at a point on the feedline is evaluated by dividing the discrete Fourier transform of voltage and current at that point. The feedline on which impedance is evaluated must be matched to the source resistance. Then, the measured impedance may be transformed to the reflection coefficient seen at the source port reference plane.

In both the FDTD simulator and in HFSS, the FR-4 substrate is modeled as a lossy dielectric, the microstrip traces are modeled as perfect electrical conductors, the SMA connectors are modeled as lumped series inductors, and the source is a lumped port with 50Ω impedance [5]. The substrate is placed in vacuum, which is bounded by a PML and infinite ground plane.

D. Fabrication and Measurement

The three MC designs are fabricated on FR-4 boards with a bottom copper ground plane, and the microstrip geometry is formed by cutting copper tape with a laser cutter. Fig. 4 shows each of the fabricated circuits. For each MC, we first connect

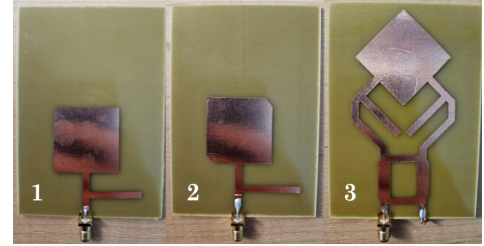


Fig. 4. The three fabricated MCs. From left to right: Circuit 1, Circuit 2, Circuit 3.

the input of the circuit to port 1 of a calibrated NanoVNA and measure S_{11} with a frequency sweep from 2 GHz to 3 GHz. Next, we also connect Circuit 1 to port 2 of the NanoVNA, and we place the two circuits such that the antennas face each other at a distance of 30 cm. We then measure S_{21} for seven different rotation angles (0° to 90°) of the port 2 antenna relative to the port 1 antenna in the plane of the circuit board.

III. LEGALITY, SAFETY AND ETHICS

A. Legality

Limits for operation in the 2.4-2.4835 MHz band are given in FCC Part 15 [12], which restricts maximum power output to 1 W and maximal gain to 6 dB (15.247.b.3). Out of band emissions must be at least 20dB below the peak gain within the band of operation (15.247.d) and power spectral density is limited to 8dBm in any 3kHz band (15.247.e) [12]. Our system operates comfortably within the above constraints.

Using the nanoVNA as a power source outputting about .1mW places us far below the power and power density

constraints. To avoid spurious emissions outside of our operation band, we take precaution to avoid harmonics within the antenna and simulate gain to ensure it is below 6 dB.

B. Safety

EM radiation exposure limits are given in FCC 1.1310.3 which sets Maximum Permissible Exposure (MPE) at 1 mW/cm^2 for this frequency range [13]. In a worst case scenario ($P=1 \text{ mW}$ $G=6 \text{ dB}$ $r=1 \text{ cm}$), using the equation given by FCC Human Exposure Guidelines $S = \frac{P \cdot G}{4 \cdot \pi \cdot R^2} = \frac{1 \text{ mW} \cdot 6 \text{ dB}}{4 \cdot \pi \cdot 1^2} = .316 \frac{\text{mW}}{\text{cm}^2}$ S is well below the MPE [14].

Fabrication safety concerns are laser cutting the FR-4 and soldering. For soldering we follow the guidelines outlined by Hunt and Thomas [15]. For laser cutting we intend to utilize the Universal Laser cutter in EDS, we will receive proper safety training at that time following institutional guidelines.

C. Ethics

Direct stakeholders for these devices include creators and users of Bluetooth products. Users would see more robustness at a cost of reduced efficiency causing more battery drainage. Engineers of Bluetooth devices would have to have more precise tolerances in their manufacturing as small errors can throw off the phase mismatch and circular polarization.

Indirect stakeholders include the environment and users of other Bluetooth technologies. The environment suffers from slightly higher EM pollution as well as higher energy usage. Users of other Bluetooth technologies may experience more interference due to 2 directions of transmission.

IV. RESULTS

The measured S_{11} results, pictured in Fig. 5, show that the three fabricated circuits exhibit resonance at about 2.2 GHz instead of the desired resonant frequency of 2.45 GHz. At the resonant frequency, $|S_{11}| < -15 \text{ dB}$ for all three antennas.

TABLE I
AXIAL RATIO OF ANTENNAS

Antenna	Axial Ratio (dB)
Circuit 1	11.1
Circuit 2	3.3
Circuit 3	4.0

We calculate the axial ratio of each device by first determining the maximum $|S_{21}|$ of circuit 1. We then subtract that by the $|S_{21}|$ achieved at a 90° offset angle. We assume the highest gain is the major axis and the minor axis is offset by 90° . We see an axial ratio of 11.1 dB for Circuit 1, 3.3 dB for circuit 2, and 4.0 dB for Circuit 3, which is depicted in Table I.

V. DISCUSSION

The results show that Circuit 2 and Circuit 3 both provide a significantly higher quality of circular polarization compared to Circuit 1, as intended. As noted previously, the resulting resonant frequency for all antennas is much lower than intended. This error can primarily be attributed to poor calibration with

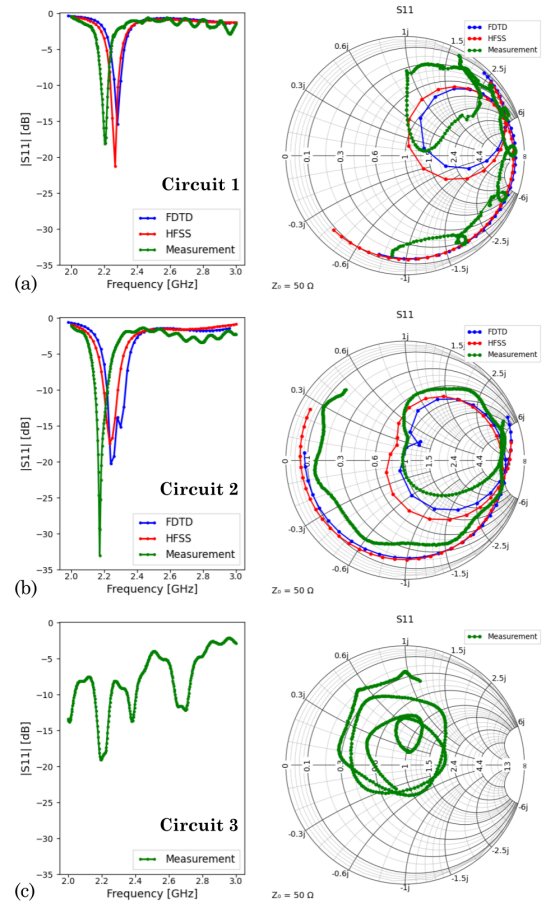


Fig. 5. S_{11} simulation and measurement for (a): Circuit 1, (b): Circuit 2, (c): Circuit 3

the substrate and microstrip launch characteristics prior to the MPA design process. By fitting closer to initial measurements, simulated and measured S_{11} are in reasonable agreement. The measured S_{11} for Circuit 3 matches expectations, as a well-designed quadrature hybrid coupler should present a matched load to the input port for any load applied to the output.

We experienced issues with inconsistent distances during angular rotation measurements. Without a testing apparatus, we had non-negligible distance differences as well as yaw and pitch differences between measurements, which likely caused errors in the axial ratio estimated. In the future, we could improve data collection by building a testing rig that isolates motion in the antenna to only the roll, keeping distance, pitch, and yaw identical while providing accurate angle readouts.

VI. CONCLUSIONS

We designed, fabricated, and evaluated the quality of circular polarization for three MPA structures targeting the 2-3 GHz range: a single-feed and a dual-feed circularly polarized MPA achieve an axial ratio of 3.3 dB and 4.0 dB, respectively, which is significantly lower than the 11.1 dB axial ratio measured for a linearly polarized design, while all designs achieve $|S_{11}| < -15 \text{ dB}$ at resonance.

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